

Bilgin

May 4th, 2020

Total points: 100

Please fill in your answers in the space supplied

Show all of your calculations, conversions and work to receive full credit.**QUESTION 1 (15 pts)****True or False (2 pts each)**

_____ The entropy in the exit of the adiabatic expansion turbine is less than the inlet of the turbine.

_____ Energy transferred by heat to a closed system during an internally reversible system can be calculated by the area under the curve in T-v diagram.

_____ Helmholtz energy is an extensive property.

_____ A closed system can experience an increase in entropy only when there is heat transfer to the system.

_____ The change in Helmholtz free energy (dpsi) is equal to zero for a phase change from a saturated liquid to a saturated vapor.

_____ The second Carnot corollary states that reversible power cycles operating between the same two thermal reservoirs have the same thermal efficiency.

Fill in the blank

(3 pts) A gas goes from State A to State B in a reversible, adiabatic process. It then goes from State B back to State A by a different pathway that is irreversible and not adiabatic. The heat transfer for the irreversible pathway is

_____ (greater than/equal to/ less than) zero. Explain.

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QUESTION 2 (10 pts)

- a) (5 pts) Water vapor enters a turbine at 10 bar $T_1 = 300^\circ\text{C}$ and exits at $p_2 = 5$ bar. The work developed is measured as 150 kJ per kg of water vapor flowing. Calculate the isentropic turbine efficiency.

- b) (5 pts) Air undergoes a process from $T_1 = 650$ K, $p_1 = 10$ bar to a final state where $s_2 = s_1$, $p_2 = 2$ bar. Employing the ideal gas model, determine the final temperature T_2 , in K. Solve using p_r data from Table A-22 and

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QUESTION 3 (10 pts)

- a) (2 pts) Write the specific Maxwell relation that corresponds to the thermodynamic function:

$g(T, P)$: _____

- b) (2 pts) Write the two, second mixed partial derivatives for the exact differential:

$dh = Tds + vdp$: _____

- b) (6 pts) A 3 kg mass of water has a pressure of 10.0 bar and temperature 150°C. What is the Helmholtz energy, in kJ?

Show all your work for full credit

QUESTION 4: (32 pts)

3 kg of water, initially at 120°C and 3bar is contained piston-cylinder device. The water undergoes a process. Heat is transferred between the piston-cylinder and a 25°C room until the water becomes a state where $x=0.6$. The piston moves freely and maintains constant pressure throughout the process. Kinetic and potential energy can be ignored. Calculate work and heat transferred in kJ. Determine if the process is possible/reversible. Carefully sketch P-v and T-s diagrams.

QUESTION 5: (33 pts)

A stream of superheated steam at 5 bar and 280 °C with a mass flow rate of 20,000 kg/min (stream 1) is mixed with a stream of liquid water at 3 bar and 20 °C (stream 2) in an adiabatic mixing chamber. The resulting mixture exits as a saturated vapor at 1 bar (stream 3). Assume steady state operation and neglect any changes in kinetic and potential energy. Calculate the mass flow rate of the liquid water entering the mixing chamber, in kg/s. Calculate the mass flow rate of the saturated vapor exiting the mixing chamber, in kg/s. Calculate the entropy production for the process, in kJ/K-s

